

React Forms

What Are Forms in React?

In React, forms behave differently than in HTML because React manages state via JavaScript. Instead of letting the DOM handle input values, we control them using React state — this is called Controlled Components.

Controlled vs Uncontrolled Components

Type	Description
Controlled	Input value is set by React state (value={state}) and updated via onChange
Uncontrolled	Input value is handled by the DOM; use ref to read values

We'll focus on Controlled Components — they're more predictable and easier to test.

Step-by-Step: Building a Controlled Form

Goal: Create a user registration form with:

- Name (text)
- Email (email)
- Age (number)
- Submit button
- Display submitted data

Step 1: Set up React State

Use useState to manage form data.

```
import React, { useState } from 'react';
import './App.css';
```

```
function App() {
  // Step 1: Initialize state for form fields
  const [formData, setFormData] = useState({
    name: "",
    email: "",
    age: ""
  });

  return (
    <div className="App">
      <h2>User Registration Form</h2>
      {/* Form will go here */}
    </div>
  );
}
```

```
export default App;
```

Step 2: Create Input Fields with value and onChange

Bind each input to its corresponding state property using `value={formData.field}` and update state on change with `onChange`.

```
<form onSubmit={handleSubmit}>
```

```
  <div>
```

```
    <label>Name:</label>
```

```
    <input
```

```
      type="text"
```

```
      name="name"
```

```
      value={formData.name}
```

```
      onChange={handleChange}
```

```
    />
```

```
  </div>
```

```
  <div>
```

```
    <label>Email:</label>
```

```
    <input
```

```
      type="email"
```

```
      name="email"
```

```
      value={formData.email}
```

```
      onChange={handleChange}
```

```
    />
```

```
  </div>
```

```
  <div>
```

```
    <label>Age:</label>
```

```
    <input
```

```
      type="number"
```

```
      name="age"
```

```
      value={formData.age}
```

```
      onChange={handleChange}
```

```
    />
```

```
  </div>
```

```
  <button type="submit">Submit</button>
```

```
</form>
```

Step 3: Handle Input Changes — `handleChange`

We write one handler that works for all inputs using the `name` attribute.

```
const handleChange = (e) => {  
  const { name, value } = e.target;  
  setFormData(prevState => ({  
    ...prevState,  
    [name]: value // Dynamic key update  
  }));  
};
```

This line `[name]: value` means:

If `name` is `"email"`, then `formData.email` gets updated.

Step 4: Handle Form Submission — `handleSubmit`

Prevent default page reload and process the form data.

```
const handleSubmit = (e) => {
  e.preventDefault(); // Prevents page refresh
  console.log('Form Submitted:', formData);
  alert(`Submitted: ${formData.name}, ${formData.email}, ${formData.age}`);
};
```

Final Complete Example

```
import React, { useState } from 'react';
import './App.css';
```

```
function App() {
  const [formData, setFormData] = useState({
    name: "",
    email: "",
    age: ""
  });

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData(prevState => ({
      ...prevState,
      [name]: value
    }));
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    console.log('Form Data:', formData);
    alert(`✔ Form Submitted!\nName: ${formData.name}\nEmail:
    ${formData.email}\nAge: ${formData.age}`);
  };

  return (
    <div className="App" style={{ padding: '20px', fontFamily: 'Arial' }}>
      <h2> User Registration Form</h2>
      <form onSubmit={handleSubmit} style={{ maxWidth: '400px', margin: 'auto' }}>
        <div style={{ marginBottom: '15px' }}>
          <label htmlFor="name">Name:</label>
          <input
            id="name"
            type="text"
            name="name"
            value={formData.name}
            onChange={handleChange}
            placeholder="Enter your name"
            required
            style={{ width: '100%', padding: '8px', marginTop: '5px' }}
          />
        </div>
      </form>
    </div>
  );
}
```

```
<div style={{ marginBottom: '15px' }}>
  <label htmlFor="email">Email:</label>
  <input
    id="email"
    type="email"
    name="email"
    value={formData.email}
    onChange={handleChange}
    placeholder="Enter your email"
    required
    style={{ width: '100%', padding: '8px', marginTop: '5px' }}
  />
</div>
```

```
<div style={{ marginBottom: '15px' }}>
  <label htmlFor="age">Age:</label>
  <input
    id="age"
    type="number"
    name="age"
    value={formData.age}
    onChange={handleChange}
    placeholder="Enter your age"
    min="1"
    max="120"
    required
    style={{ width: '100%', padding: '8px', marginTop: '5px' }}
  />
</div>
```

```
<button
  type="submit"
  style={{
    backgroundColor: '#007bff',
    color: 'white',
    border: 'none',
    padding: '10px 20px',
    cursor: 'pointer',
    fontSize: '16px'
  }}
>
  Submit
</button>
</form>
```

```
{/* Optional: Display submitted data */}
{Object.keys(formData).length > 0 && (
  <div style={{ marginTop: '20px', padding: '10px', backgroundColor: '#f0f0f0' }}>
    <h3>Current Form Data:</h3>
```

```
        <pre>{JSON.stringify(formData, null, 2)}</pre>
      </div>
    )}
  </div>
);
}

export default App;
```